

* A theory is renormalizable if all coeff. have positive or zero mass dim

• look at ϕ^3 in $d=6$

determine how to construct a finite expression for a scattering amp. to arbitrarily high order in the ϕ^3 coupling g .

$$\sum \text{---} \textcircled{1PI} \text{---} = i\pi(k^2)$$

which gives $i\pi(k^2)$. Then we sum all 1PI diagrams at V_3

$$\sum \textcircled{1PI} = iV_3(k_1, k_2, k_3)$$

• Order by order we adjust the values of Z_ϕ, Z_m, Z_g to maintain

$$\pi(-m^2) = 0$$

$$\pi'(-m^2) = 0$$

$$V_3(0, 0, 0) = g$$

* n -pt vertex $V_n(k_1, \dots, k_n)$
 $4 \leq n \leq E$

• Compute skeleton expansion

Def: draw all the contributing 1PI diagrams but omit diagrams that include either a prop. or 3pt vertex correction.

• i.e. omit any 1PI diagrams that contain a subdiagram w/ 2 or 3 external lines